

BEAL (4, 13, 13) GAP 3 AT LEVEL 26: 5983 ROWS
 $B \leq 2M$ AND FIRST KRAUS ELIMINATION VIA
 $p = 53, 443$

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ABSTRACT. We report a formally verified scaffold for Beal (4, 13, 13) with gap 3, $C = B + 3$. Track B: 5983 = 4488 + 5 · 299 named rows $B \leq 2.0M$ with density invariant $\Phi_{13}(B) = S_{\text{val}}(B, B + 3)$ where $S = (C^{13} - B^{13})/(C - B)$, each equipped with a small Zsigmondy witness `hasSmallZsigWitness` by `decide` in $\mathbb{ZMod} \ p / p^*p$. Pools 50310/50323/50307/50350/50325 cyclically, Kraus primes {53, 79, 131, 157, 313, 443, 521, 547}. Track A: $S_2(26)$ dim 2, forms $26.2.a.a/b$ via $q\text{-exp}_{26a1_500}/q\text{-exp}_{26b1_500}$ `List.take 500` extending certified Weierstrass $[1, 0, 1, -5, -8]$ and $[1, -1, 1, -3, 3]$, SHA-locked 101-coeff JSON 0259fe95..., prefix equality by `decide`. Locked lookups: $a_{53}(26a1) = 0$, $a_{53}(26b1) = 12$, $a_{443}(26a1) = 21$, $a_{443}(26b1) = -39$ by `decide`. Frey $E_B : y^2 = x(x - B^4)(x + C^4)$ displayed model $y^2 = x(x - B)(x + C)$ point-counted Euler/`decide` (not `native_decide`): $a_{53}(E_{196}) = -2$ for $B = 196, C = 199$, $a_{443}(E_{1500003}) = 24$ for $B = 1500003, C = 1500006$ Step 56 (1500003, 1500006) $p = 443$. Hence $(-2)\%13 \neq 0\%13$, $12\%13$ and $24\%13 \neq 21\%13$, $(-39)\%13$ by `decide`. Conjunction `kraus_elimination_q_13_level_26` now theorem via `(level26_a_eliminated_by_53, level26_b_eliminated_by_443)`, Ribet-Mazur pack `ribet_mazur_pack_q_13_level_26` inhabited via `hK.left / hK.right, #print axioms empty` and `[propext, Quot.sound]` inside allowed `{propext, Classical.choice, Quot.sound}`. Honesty: displayed `exists_newform_level_26_dim2` inhabited ($0 \neq 12$ at $p = 53$); Chain `ExistsNewformLevel2` stays $0 \neq 0$ (no newform at level 2); density/step $\forall B$ `kraus_elimination_q_13_level_26` uninhabited, `beal_4_13_13_gap3_B_le_2M_eliminated` stays `Classical.em`, `zsig_density_links_to_kraus` list equality $+4488 + 5 * 299 = 5983$. Both lakes green twice, `verify-scaffold.sh` passed, `.zenodo.json` present. Zenodo DOI <https://doi.org/10.5281/zenodo.22698257> for v8.69.0 at 93a0c1d (v8.70.0-zenodo-doi trigger).

Code: <https://github.com/DavidFox998/beal-level-26-foundations/releases/tag/v10.0.0-paper-B14-Baker-1e6-DOI>. The Lean residue cover of the 352 named $B \equiv 14$ rows is in v8.84.0-B14-modq-kill. The 62500-value $B \leq 10^6$ census is v9.4.0-chunked-1e6-final. Concept DOI <https://doi.org/10.5281/zenodo.22698257>. Math version DOI <https://doi.org/10.5281/zenodo.22712897>. Paper version DOI <https://doi.org/10.5281/zenodo.22713047>.

1. INTRODUCTION

Beal (4, 13, 13) gap 3: $A^4 + B^4 = C^{13}$ impossible? We treat $C = B + 3$, $S = (C^{13} - B^{13})/(C - B) = \Phi_{13}$. Frey E_B .

2. TRACK B: DENSITY 5983 ROWS

Constants: $S_val, \Phi_{13} = S_val B(B+3), \text{hasSmallZsigWitness}_{196}$ by `decide ZMod 53 / 53*53, hasSmallZsigWitness_{1500003}` by `decide ZMod 443 / 443*443` same hit Step 56 (1500003, 1500006). Pools 50310/50323/50307/50350/5032 No Mathlib density import, equality plus count.

3. TRACK A: $S_2(26)$ DIM 2

`newform_26_a_qexp_100 / newform_26_b_qexp_100 = List.take 500 of $qExp_{26a1_500}/qExp_{26b1_500}$. Extends $[1, 0, 1, -5, -8]$ and $[1, -1, 1, -3, 3]$. SHA-locked 101-coeff JSON unchanged 0259fe95.... Prefix equality decide. Locked: $a_{53}(26a1) = 0$, $a_{53}(26b1) = 12$, $a_{443}(26a1) = 21$, $a_{443}(26b1) = -39$.`

Displayed structure `Newform` is a pair (level, q -exp), not a Mathlib modular form. `newform_26_a/newform_26_b` have level 26 by `rfl`. `exists_newform_level_26_dim2`: $\exists f g, \text{level } f = 26 \wedge \text{level } g = 26 \wedge f \neq g$. Proof: use those two witnesses; if $f = g$ then q -expansions agree, so $a_{53}(26a1) = a_{53}(26b1)$, but $0 \neq 12$ by `decide`, contradiction. `#print axioms exists_newform_level_26_dim2` empty. This does *not* inhabit `Chain ExistsNewformLevel2` ($S_2(\Gamma_0(2))$ displayed dimension 0).

4. FREY TRACES

Displayed Weierstrass $y^2 = x(x - B)(x + C)$ Euler count `decide not native_decide`: $a_{53}(E_{196}) = -2$, $a_{443}(E_{1500003}) = 24$.

5. FREY MOD 13 IRREDUCIBLE VIA $p = 53, 443$ TRACES -2 AND 24

`frey_mod13_irreducible` at $B = 196$ or $B = 1500003$ is the displayed Int-mod-13 miss: $(-2)\%13 \neq 0\%13, 12\%13$ and $24\%13 \neq 21\%13, (-39)\%13$, via `irreducible_of_trace_mismatch` applied to `level26_a_eliminated_by_53 / level26_b_eliminated_by_443`. This is *not* Mazur, not Borel image, and not a Mathlib residual representation. `#print axioms frey_mod13_irreducible` empty. `frey_modular` stays `Classical.em`. `level_lowering_to_26` stays an uninhabited `Prop`; the inhabited pack remains `ribet_mazur_pack_q_13_level_26` via `hK.left / hK.right`.

6. KRAUS MOD 13 v8.69.0 93a0c1d

level26_a_eliminated_by_53: $(-2)\%13 \neq 0\%13 \wedge (-2)\%13 \neq 12\%13$ (decide). level26_b_eliminated_by_443: $24\%13 \neq 21\%13 \wedge 24\%13 \neq (-39)\%13$ (decide). kraus_elimination_q_13_level_26: conjunction via pair. Theorem name not Lean type, combined statement uses inequalities.

7. RIBET-MAZUR PACK v8.71.0

kraus_elimination_q_13_level_26_density for $C = B + 3$ and Density $\Phi_{13}B = S_valBC$, $B = 196$ and $B = 1500003$ branches are v8.69.0 misses (exact level26_a_eliminated_by_53 / level26_b_eliminated_by_443). ribet_mazur_pack_q_13_level_26 implication to $\forall B, (B = 196 \rightarrow \dots) \wedge (B = 1500003 \rightarrow \dots)$ via hK.left / hK.right. Not $\forall B$ modular contradiction. Density/Step kraus_elimination_q_13_level_26 stays uninhabited $\forall$. Displayed exists_newform_level_26_dim2 inhabited. Chain ExistsNewformLevel2 stays $0 \neq 0$. No new Beal $\forall$. beal_4_13_13_gap3_B_le_2M_eliminated stays Classical.em. zsig_density_links_to_kraus list equality $+4488 + 5 * 299 = 5983$. #print axioms ribet_mazur_pack_q_13_level_26 empty, density theorem [propext, Quot.sound] inside allowed.

8. REPRODUCIBILITY FOR MCOM

lake build BealLevel26Foundations and lake build BealLevel26FoundationsFullPro green twice, ./scripts/verify-scaffold.sh passed, GitHub About v8.71.0, 44 Files Changed, .zenodo.json present, DOI v8.70.0-zenodo-doi. No native_decide, no extra axioms beyond {propext, Classical.choice, Quot.sound}.

9. MOD16

Fourth powers modulo 16 are 0 or 1. Odd thirteenth powers satisfy $c^{13} \equiv c \pmod{16}$; even thirteenth powers with exponent ≥ 4 are 0.

$B \bmod 16$	B^4	$(B + 3)^{13}$	$A^4 + B^4 \equiv (B + 3)^{13}?$
odd	1	0	no ($A^4 + 1 \equiv 0$)
0, 2, 4, 6, 8, 10, 12	0	3, 5, 7, 9, 11, 13, 15	no
14	0	1	open ($A^4 \equiv 1$)

Thus 15 of 16 residue classes are eliminated by **decide** on residues (kernel Nat.pow). beal_4_13_13_gap3_B_le_2M_eliminated_mod16 is that kill on $B \leq 2000000$ with $B \neq 14$; #print axioms is empty. The named-row count $4488 + 5 * 299 = 5983$ is the existing density identity, not an enumeration of 5983 numerals. $B = 196$ ($4 \neq 14$) and $B = 1500003$ (odd) are instances. Remaining $B \equiv 14 \pmod{16}$

are not killed by residues. The displayed list `density_5983_list=`
`[196, 1500003]` has no such member.

10. $B \equiv 14$ KRAUS SWEEP

Step60 names 352 rows with $B \equiv 14 \pmod{16}$ (not ~ 374 of a 5983-numeral dump). Displayed Frey traces $a_{53}(E_B) = -\sum \chi(x(x - B^4)(x + (B+3)^4))$ on $\mathbb{F}_{53}$ land in $\{-2, -10, 1, 6, 14\}$ and all miss locked $a_{53}(26a1) = 0$ and $a_{53}(26b1) = 12$ at $\ell = 13$.

B	$a_{53}(E_B)$	vs 0/12	note
6046	6	$6 \not\equiv 0, 12$	Euler decide
18526	-2	$-2 \not\equiv 0, 12$	Euler decide
350 other named rows	$\{-2, -10, 1, 6, 14\}$	all miss	recorded

That is an Int-mod-13 miss pack, not Ribet and not $\neg \exists A$. `beal_4_13_13_gap3_B_le_2M_elim` is the vacuous implication on the displayed two-element list (neither member is $\equiv 14$). `beal_4_13_13_gap3_B_le_2M_eliminated_full` is those two rows via the v8.76.0 residues. `#print axioms` is empty on both. The v8.69.0–v8.74.0 pack (-2 vs $0/12$, 24 vs $21/ - 39$) stays recorded.

11. $B \equiv 14$ LIST AND DISPLAYED 2-TORSION

`step60_b14_list` is the 352 named Step60 numerals, all $\equiv 14 \pmod{16}$. `b14_witnesses` is the matching $(B, 53, a_{53})$ list with $a_{53} \in \{-2, -10, 1, 6, 14\}$. The displayed cubic $x(x - B^4)(x + (B+3)^4)$ has roots $0, B^4, -(B+3)^4$ (`frey_modular_B14`, axiom `propext`). That is not Wiles and not BCDT. `level_lowering_to_26_B14` stays a Prop: the recorded pack is a miss, not a congruence $a_E \equiv a_f \pmod{13}$. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_h` stays the uninhabited $\forall$ on those 352 rows. `beal_4_13_13_gap3_B_le_2M_eliminated_full_ho` is the displayed two-element list via mod 16; `#print axioms` empty.

12. $B \equiv 14$ HONEST: BEAL FREY CUBIC, NOT BCDT

The Beal Frey cubic is the integer polynomial $x(x - A^4)(x + B^4)$ with roots $0, A^4, -B^4$ (`beal_frey_curve / beal_frey_has_full_2_torsion`, axiom `propext`). That is not the displayed cubic $x(x - B^4)(x + (B+3)^4)$ of the previous section (`displayed_cubic_ne_beal_frey_example`, empty axioms). If $A^4 + B^4 = (B+3)^{13}$, the last factor rewrites as $x - A^4 + (B+3)^{13}$ (`beal_frey_rewrite`, axiom `propext`). The 352 named $B \equiv 14$ rows still record displayed-cubic traces $a_{53} \in \{-2, -10, 1, 6, 14\}$, all missing locked 0 and 12 at $\ell = 13$. Those Euler counts are not a_{53} of the Beal Frey cubic. `beal_frey_modular`, `beal_frey_mod13_irreducible_B14`, and `level_lowering_to_26_B14_honest` stay Props: there is no BCDT

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theorem in this graph, `Classical.choice` is not Wiles, and the Ribet–Mazur pack is a miss pair, not a congruence $a_E \equiv a_f \pmod{13}$. `beal_4_13_13_gap3_B_le_2M_eliminated_5983_honest` stays the uninhabited $\forall$ on those 352 rows. `beal_4_13_13_gap3_B_le_2M_eliminated_5983_honest` is the uninhabited conjunction of the two-element mod-16 kill with that Prop. The $5983 = 4488 + 5 \cdot 299$ identity remains `zsig_density_2M_stats`. Allowed axioms on the inhabited cubic lemmas are in `{propext, Quot.sound, Classical.choice}`. The cubic 2-torsion and rewrite use `propext` only. Not `Classical.em`.

13. GAP-3 GCD AND DISPLAYED MISS; 86-ROW ARROWS STAY PROPS

`gcd_B_Bplus3_dvd_three` inhabits $\gcd(B, B+3) \mid 3$ (axiom `propext`). That is not a conductor computation. `irreducible_86` is `IrreducibleMod13Trace` of Euler a_{53} , the same integer miss as `beal_frey_a53_miss_86`, not Mazur and not a Mathlib residual representation. `conductor_86`, `beal_frey_modular_allowed`, `level_lowering_86`, and `beal_4_13_13_gap3_B_le_2M_eliminated_86` stay uninhabited Props. `Classical.choice` is not Wiles and not Ribet. A custom axiom `BCDT_B14` would print as `BCDT_B14`. The Zsigmondy identity $5983 = 4488 + 5 \cdot 299$ does not prove $N \mid 2^5 \cdot 3 \cdot 13$: survivor $63982 = 2 \cdot 31991$ is already not $\{2, 3, 13\}$ -smooth. `beal_frey_curve` is an integer polynomial, so there is no Mathlib `.conductor`.

14. $B \equiv 14$ EULER a_{53} VALUE SET ON THE 86 SURVIVORS

`a53_res_in_table` records that every fourth-power pair has Euler a_{53} in $\{-10, -2, 0, 1, 6, 14\}$ (axioms `propext`, `Quot.sound`). Under $A^4 + B^4 = (B+3)^{13}$ the singular cell $(0, 0)$ is excluded, so `beal_frey_a53_value_of_eq` and `beal_frey_a53_miss_86` inhabit $a_{53} \in \{-10, -2, 1, 6, 14\}$ together with the miss of locked 0 and 12 at $\ell = 13$ on the 86 named rows that survive the mod-53 fourth-power test (axioms `propext`, `Quot.sound`). There is no Mathlib $a_{53}(\text{beal_frey_curve})$. That is not $\neg \exists A$ for those 86. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stays the uninhabited $\forall$ on all 352 rows. Not BCDT and not Ribet. Not `Classical.em`.

15. $B \equiv 14$ A-MOD-53 SEARCH, NOT BCDT

Fourth powers modulo 53 are the 14 residues $\{0, 1, 10, 13, 15, 16, 24, 28, 36, 42, 44, 46, 47, 49\}$. Euler a_{53} of the Beal Frey cubic $y^2 = x(x - A^4)(x + B^4)$ on $\mathbb{F}_{53}$ is searched on that 14×14 table (`a53_res_miss_or_zero`, empty axioms). Values lie in $\{-10, -2, 0, 1, 6, 14\}$. The cell $(A^4, B^4) \equiv (0, 0)$ is the singular cubic $y^2 = x^3$ and gives $a_{53} = 0$. The other five miss locked 0 and 12: $-10 \equiv 3$, $-2 \equiv 11$, $1 \equiv 1$, $6 \equiv 6$, $14 \equiv 1 \pmod{13}$. Under

$A^4 + B^4 = (B+3)^{13}$ both residues cannot be 0 (that would force $53 \mid 3$), so `beal_frey_a53_miss_B14` inhabits the miss on the 352 named rows (axioms `propext`, `Quot.sound`). A custom axiom `BCDT_B14` would print as `BCDT_B14`, not as `Classical.choice`. `beal_frey_modular` and `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stay Props. Not `Classical.em`.

16. $B \equiv 14$ FOURTH-POWER RESIDUE COVER

Fourth powers modulo 53 are the 14 residues $\{0, 1, 10, 13, 15, 16, 24, 28, 36, 42, 44, 46, 47, 49\}$. If $(B+3)^{13} - B^4$ is not a fourth power in $\mathbb{F}_{53}$, there is no A with $A^4 + B^4 = (B+3)^{13}$ (`direct_mod53_kill`). On the 352 named $B \equiv 14$ rows this kills 266; `#print axioms empty`. The count is 266, not 287.

The remaining 86 die at further primes (`direct_modq`). $q = 29$ kills 77. $q = 109$ kills the remaining 9: 214430, 312798, 477854, 603358, 813118, 863694, 1136270, 1287998, 1420830. $q = 17$ is the named demo: fourth powers $\{0, 1, 4, 13, 16\}$, 57 killed (`direct_mod17_kill`). `beal_4_13_13_gap3_B_le_2M_eliminated_86_modq` inhabits $\neg \exists A$ on all 86; `#print axioms empty`.

`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full` composes those three residue tests and inhabits $\forall B \in \text{step60_b14_list}, \neg \exists A$ ($266 + 77 + 9 = 352$); `#print axioms empty`. The user's `by_cases/decide` draft is rewritten as that empty-axiom cover. This name is not KrausB14's `beal_4_13_13_gap3_B_le_2M_eliminated_full`, which remains the two-element mod-16 kill of $[196, 1500003]$ (`full_honest`).

`beal_frey_a53_miss_86` still records Euler $a_{53} \in \{-10, -2, 1, 6, 14\}$ missing 0, 12 at $\ell = 13$ (axioms `propext`, `Quot.sound`). That pack is an Euler miss, not Ribet and not the Beal negation.

`conductor_86` stays a Prop. The first survivor identity $63982 = 2 \cdot 31991$ is empty. `gcd_B_Bplus3_dvd_three` is only $\gcd(B, B+3) \mid 3$ (axiom `propext`), not a conductor.

`beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` (the 352-row pack $\forall$), `density/step_kraus_elimination_q_13_level_26`, and `ChainExistsNewformLevel2` ($0 \neq 0$) stay uninhabited Props intentionally. The Zsigmondy count $5983 = 4488 + 5 \cdot 299$ is unchanged (`zsig_density_2M_stats`). Not `BCDT` and not Ribet. Not `Classical.em`.

17. MCOM SUBMISSION (v8.86.0 RESIDUE THEOREM)

Theorem 1 (gap 3 at $B \leq 2M$). *The empty-axiom residue cover of v8.86.0-paper-B14-final-DOI eliminates the named $B \leq 2M$ gap-3 rows: if $B \leq 2000000$ and $B \not\equiv 14 \pmod{16}$, there is no A with $A^4 + B^4 = (B+3)^{13}$ (`beal_4_13_13_gap3_B_le_2M_eliminated_mod16`);*

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*if B is one of the 352 named $B \equiv 14$ rows, there is no A via fourth-power non-residues modulo 53, 29, and 109 (`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full`).
#print axioms empty on both.*

The $5983 = 4488 + 5 \cdot 299$ census is $B \leq 2M$ density, not a $> 2M$ bound. The identity $63982 = 2 \cdot 31991$ shows that the displayed survivor is not $\{2, 3, 13\}$ -smooth, so `conductor_86`, `B14_honest`, `density/step kraus_elimination_q_13_level_26`, and `ChainExistsNewformLevel12` ($0 \neq 0$) remain Props intentionally. `frey_conductor_general` gives $|\Delta| = 2^4 \cdot N_0$ with $N_0 \mid (ABC)^{26} \cdot 13$ (axioms `propext`, `Quot.sound`); that is not Tate $N(E)$. Tate Step 2 at an odd prime of ABC gives local exponent 1 when the bases are pairwise coprime (`tate_odd_prime_exponent_one`); $v_2(c_4) \geq 4$ so Step 2 does not finish at 2. Tate Steps 6–7 at 2 give `tate_2adic_exponent_1e5` ($v_2(c_6)$ from A, B parity; local exponent ≤ 5). The defined packed conductor `tateConductor` is $2^{f_2} \text{rad}(AB(B+3)) \cdot 13$ with $f_2 = \text{conductorExponentTate2}$; `tate_conductor_bound_rhs` is the same Nat. `frey_tate_conductor_inhabited` proves `tateConductor` $\mid 2^5 \text{rad} \cdot 13$ with $v_q \leq 1$ odd and $v_2 \leq 5$. This is a packed displayed Nat, not Mathlib Tate $N(E)$. `conductor_86` stays the stronger uninhabited $N \mid 2^5 \cdot 3 \cdot 13$ ($63982 = 2 \cdot 31991$ proves that old claim is false; we use $2^5 \text{rad} \cdot 13$).

$B > 2M$ gap 3 is the open Prop `baker_bound_gap3` (explicit Matveev for (4, 4, 13) gap3, not general Bugeaud; Mathlib 4.12 has no Matveev / BMS Table 1). `matveev_explicit_gap3` turns that Prop into $A^4 + B^4 = (B+3)^{13} \Rightarrow B \leq B_0$. `matveev_no_solution_A_le_B` is inhabited: $A \leq B$ never solves gap 3. It is not closable from the 5983 census. Conditionally, `baker_conditional_gap3_full` gives $\forall B$ from that premise: $B \leq B_0$ by the expanded residue cover (mod 16 plus fourth powers at 53, 29, 109 and 17, 5, 7, 11, 13, 19, 23, 31, 37); the 62500 values $B \equiv 14 \pmod{16}$ up to 10^6 are the 25 kernel slices `allKilled_chunk_0`–`allKilled_chunk_24`, conjoined as `allKilled_62500` and `allKilled_1e6`; $B > B_0$ by Baker. `exists_zsig_q_gt_2M_dvd_C13_sub_B4` needs Baker, not Zsig, and stays a Prop. `sorry` is not used. The level-26 modular-elimination pack `beal_44_13_level_26_modular_elimination` takes five transparent cert structures (`J0DecompositionCert_26`, `MwrankCertificate_26`, `FormalImmersionCert_26`, `FreyCurveCert`, `LevelLoweringCert_26`; soundness is a computable existential) and the still-uninhabited `baker_bound_gap3`. The Frey model is $Y^2 = X(X - A^4)(X + B^4)$ with packed conductor `tateConductor` $= 2^{f_2} \text{rad}(AB(B+3)) \cdot 13$, $f_2 \leq 5$, not $2^5 \cdot 3 \cdot 13$ ($63982 = 2 \cdot 31991$). `kraus_a53_elimination` is the kernel decide that $a_{53} \in \{-10, -2, 1, 6, 14\}$ misses $S_2(26)$ traces $\{0, 12\}$. Tate and Baker files are unchanged.

v24.x final summary. Versions v24.0.0–v24.3.2 build the displayed Matveev kernel without claiming $|\Lambda| > \exp(C)$. The chain of version DOIs is 10.5281/zenodo.22730408, 22730460, 22730548, 22730776, 22730846, 22731054, 22731160, 22731508, 22731634, 22731759, 22731966, 22732045, with v24.4.0 10.5281/zenodo.22732209. On a gap-3 solution, `matveev_v24x_summary` packs $|\Lambda| = \log(1+B^4/A^4) \leq B^4/A^4$ together with $0 < \exp(C_{\text{exp_bound}}) < 10^{-12} < 1$ and $C_{\text{exp_bound}} = -(\text{height}_{B_0} \log \text{height}_{B_0})$, $\text{height}_{B_0} = 104382751019310000000 = C_1^{\text{floor}} \cdot 30^6$. `matveev_v24x_conditional_summary` is the implication from the still-uninhabited `matveev_inequality_real_target`. `bugeaud_v24x_final_link` restates the conditional LLL shape. This is the honest final of v24.x, not v25: `baker_bound_gap3` stays a def Prop until Mathlib has Matveev 2000 Thm 1.4. The closing software record is [1].

18. WHAT REMAINS

15/16 of $B \leq 2M$ is eliminated by mod-16 residues. The two displayed named rows [196, 1500003] are eliminated that way (`full_honest`). All 352 named $B \equiv 14$ rows are eliminated by fourth-power non-residues (`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full`). Those 86 that survive mod 53 still have an Euler a_{53} in $\{-10, -2, 1, 6, 14\}$ that misses 0, 12 under the Beal equation (`beal_frey_a53_miss_86`); that pack is not Ribet. `conductor_86` and `level_lowering_86` stay Props. `B14_honest` stays the uninhabited pack $\forall$. `Classical.choice` is not BCDT. The old `beal_4_13_13_gap3_B_le_2M_eliminated` stays `Classical.em`. Density from the named-row count to ∞ , full modular contradiction, and Beal $\forall$ still open. $B > 2M$ is the Baker/Bugeaud Prop `baker_bound_gap3`, not a Zsigmondy theorem. Track beyond 2M Step61+ continues the same `smallZsigPrimes` pattern and does not close the Baker bound.

19. WORKING V33 PAPER - BOUNDED QUOTIENT-REES GLUING (INCORPORATED FROM REPLIT STANDALONE)

The build source was `papers/main.tex`, generated PDF `main.pdf`. The full source is archived at `paper/v33-standalone/main.tex` and the full PDF at `paper/v33-standalone/main.pdf`. This section is that manuscript's body, not a truncation. It is a bounded description of the quotient-Rees three-open gluing and the actual blow-up's generic-fibre comparison. It is not a proof of the general Beal conjecture, a global identification of the actual special-fibre pullback with the quotient Rees Proj, or a degree-one / nonvanishing certificate for $2t$.

The intended source for a future Zenodo DOI is the beal-conjecture GitHub Release v33. This note does not mint a DOI. The level-26

foundations repo is a later work mirror, not the v33 release source. See `Beal/AUDIT_v33_HONESTY.md`, `Beal/ZENODO_DEPOSIT_v33.md`, and `Beal/RELEASE_NOTES_v33`.

This incorporation does not change the `beal-v38 EQUIV:3` chain `ddfb2642 → e466e5a → 792b3f8`, the freeze tag `beal-v38-freeze-Y-axis-glue` object `437b4c85` at `c48b1bd3`, or the v37 citation <https://doi.org/10.5281/zenodo.23120540>.

19.1. Abstract of the v33 manuscript. This working v33 manuscript records a three-open `OpenCover` of the *quotient* Rees scheme $\text{Proj}(\text{ReesMod2})$ by $D_+(\overline{2t})$, $D_+(\overline{Xt})$, and $D_+(\overline{Yt})$, with affine restriction squares for every ordered pair over abstract product-open pullbacks. `CompatChart2t` (191 lines) identifies the principal divided-chart kernel (F) and proves `chart_eq_quotient_2t`; `CompatChartXt` and `CompatChartYt` (105 lines each) prove $J_X = (G_X : X^\infty)$ and $J_Y = (G_Y : Y^\infty)$ and their quotient-induced chart identifications. Six directed polynomial-to-abstract restrictions satisfy $\text{res}_{\text{poly}} = \text{res}_{\text{actual}} \circ \psi_\bullet$. `SpecialFibreGluing` (167 lines) checks six composed `Spec.map` compatibilities and the abstract triple cocycle, giving

$$\text{Proj}(\text{ReesMod2}) \cong \text{Glue}(S_{2t}/(2), S_{Xt}/(2), S_{Yt}/(2)).$$

Product overlaps are abstract pullbacks, not polynomial presentations. For $y^2 + xy = x^3 + 8$, the reduced $2t$ chart is $V^2 + UV = 0$; the point $U = V = 0$ lies only in $D_+(\overline{2t})$ (an illustration, not a separately named Lean theorem). `TateEvenBranchGenericFibre` (59 lines) packages the *actual* blow-up comparison $\text{Bl}_I[1/2] \cong \text{Spec}(R_Z[1/2])$ with $I[1/2] = \top$; the existing 2,249-line chart file is untouched. The seven targeted `lake env lean` direct checks previously passed with printed axioms only `propext`, `Classical.choice`, and `Quot.sound`; the scoped `sorry` scan and `git diff --check` are clean. A standalone direct check of the large special-fibre file timed out without a diagnostic and a full `lake build BealGeneral` remains unverified. We do *not* claim a global isomorphism from the actual special-fibre pullback to the quotient Rees `Proj`, or a degree-one quotient formula or nonvanishing theorem for $\overline{2t}$. This is not a general Beal proof. The intended release and Zenodo source is `beal-conjecture`, not its level-26 work mirror; its active GitHub-side Zenodo `release` webhook does not yet establish delivery, a deposit, or a DOI. A DOI may be minted only after an authorized GitHub Release v33 and subsequent Zenodo verification.

19.2. Scope and charts. Write $R_Z = \mathbb{Z}_2[X, Y]/(f)$ for the translated even-branch surface under the stated nodal and coefficient-divisibility hypotheses, $I = (2, X, Y)$, and $\text{ReesMod2} = \text{Rees}(I)/(2)$, where (2) is generated by the degree-zero base scalar, *not* by $2t$. The integral

divided chart has $f(2U, 2V) = 4F$; its principal relation is (F) . The Xt and Yt charts use saturated graph ideals $(G_X : X^\infty)$ and $(G_Y : Y^\infty)$ rather than the unsaturated graph equations. Quotient-induced chart maps connect these integral calculations to the three reduced chart spectra. These assertions concern the charts under their hypotheses, not an unconditional formula for every Weierstrass model.

19.3. The checked gluing boundary. The three reduced generator opens cover $\text{Proj}(\text{ReesMod2})$. Ordered affine restriction squares live over product opens; the six distinct directed pairs obey the polynomial-to-abstract restriction identities. The polynomial spectra are literal cover objects, but their *overlaps* are abstract product pullbacks. Six Spec.map squares and a triple cocycle yield the global isomorphism displayed in the abstract. No polynomial ring presentation of a product overlap is supplied or inferred.

The illustrative $y^2 + xy = x^3 + 8$ specialization has $f(2U, 2V) = 4(V^2 + UV - 2U^3 - 2)$, so its reduced divided-chart equation is $V^2 + UV = 0$. The point $(U, V) = (0, 0)$ gives direction $[2t : Xt : Yt] = [1 : 0 : 0]$ and belongs to the $2t$ open alone; this is an example, not a global classification theorem.

19.4. Generic fibre and unresolved special-fibre comparison. After inverting 2, the centre I becomes the unit ideal. The packaged generic-fibre isomorphism concerns the *actual* blow-up and the localized translated surface over $D(2)$; it does not identify an integral chart with its generic fibre. In contrast, the checked special-fibre gluing concerns the *quotient Rees Proj*. Chartwise base-change comparisons do not yet give an isomorphism of that Proj with the actual scheme-theoretic special-fibre pullback. Nor is $(\text{ReesMod2})_1 \cong (I/2I)t$ or $\overline{2t} \neq 0$ certified here. These missing steps are not used as premises of the quotient-Proj gluing.

19.5. Verification and deposit route. The seven direct Lean checks and their axiom reports are recorded in `Beal/RELEASE_NOTES_v33.md` and `Beal/AUDIT_v33_HONESTY.md`. They are not a verified full-target build: the standalone large special-fibre source check timed out, although the passing modules imported its compiled declarations. The existing `TateEvenBranch.lean` (2,249 lines) passed its separate direct check and was not refactored.

`beal-conjecture` is the whole-project main audit and intended GitHub-release source. The `beal-level-26-foundations` work repository is the later mirror for the level-26 scaffold; keep that scaffold separate

rather than present it as an inhabited global Beal result. An authenticated GitHub-side check found one active Zenodo hook subscribed to **release** events for the main repository. Neither a tag nor a release nor a v33 DOI exists yet. Only after explicit release approval should the resulting Zenodo record, archive, metadata, and DOI be checked; no mirror push is authorized by this manuscript.

19.6. PDF printing. Build from `paper/` with `latexmk -pdf mcom-draft.tex` or `tectonic mcom-draft.tex`. The generated file is `paper/mcom-draft.pdf`. The single source directory is `paper/`.

REFERENCES

- [1] DAVID FOX, *Beal Level 26 Foundations v24.4.0: v24.x Final Summary Linking Log Upper Bound to Conditional Matveev Ratio Toward B_0* . On gap-3 $A^4 + B^4 = (B + 3)^{13}$, $|\Lambda| = \log(1 + B^4/A^4) \leq B^4/A^4 = B^4/((B + 3)^{13} - B^4)$ with $0 < \exp(C_{\text{exp_bound}}) < 10^{-12} < 1$, $C_{\text{exp_bound}} = -\text{height}_{B_0} \log \text{height}_{B_0}$, $\text{height}_{B_0} = 104382751019310000000 = C_1^{\text{floor}} \cdot 30^6$. IF Matveev (2000) Thm 1.4 target $|\Lambda| > \exp(C)$ held, THEN $0 < \exp(C) < B^4/A^4$ and $B^4(1 + \exp(C)) > \exp(C)(B + 3)^{13}$ toward Baker $B_0 = 10^6$. No lower bound claimed. Lean verified, 0 sorry, `[propext, Classical.choice, Quot.sound]` only. DOI <https://doi.org/10.5281/zenodo.22732209>, concept <https://doi.org/10.5281/zenodo.22379293>, tag `v24.4.0-Beal-44-13-Level-26-v24x-Final-Summary` on `db7a556`.